How Do Evidence Ordering and Conflicting Contexts Affect Answer-Abstention Decisions in Large Language Models?

Literature Review

REVIEW OF RELATED STUDIES

Xie et al. (2024) investigated large language model (LLM) behavior under knowledge conflicts, revealing that LLMs are highly receptive to single coherent counter-memories but exhibit a strong confirmation bias toward parametric memory when multiple conflicting sources are present (Xie et al., 2024, 1). Building upon how models handle conflicting knowledge states, Wen et al. (2025) broadened this perspective by conducting a comprehensive survey on abstention mechanisms in large language models, organizing the literature around a tripartite framework comprising query answerability, model knowledge, and human values (Wen et al., 2025, 2). A key strength of the survey is its holistic mapping of abstention strategies across the entire model lifecycle (pretraining, alignment, and inference), although its broad scope lacks deep empirical validation of specific domain applications (Wen et al., 2025, 2).

While broad surveys outline general abstention taxonomies, addressing specific semantic ambiguities in retrieval settings remains crucial. Addressing this, Patel et al. (2026) introduced RAAD, a retrieval-augmented framework that formulates question ambiguity detection as an answer-centric inference problem by evaluating semantic incompatibility across candidate answers (Patel et al., 2026, 3). A key strength of the study is its strong cross-dataset generalization using a lightweight cross-encoder, though its effectiveness is limited by reliance on answer plurality and sensitivity to early retrieval noise (Patel et al., 2026, 3).

Extending the challenge of ambiguous evidence to high-stakes vertical domains, Kim et al. (2025) evaluated eleven foundation models across seven medical reasoning and information retrieval tasks, identifying medical hallucinations primarily as reasoning-driven failure modes rooted in causal and temporal errors rather than mere knowledge deficits (Kim, 2025, 4). A major strength of the study lies in its multi-tiered evaluation combining quantitative benchmarks with expert physician audits and clinician surveys, although it is limited by moderate inter-rater agreement inherent in subjective clinical evaluations (Kim, 2025, 4).

To more rigorously operationalize selective abstention in scientific claim verification, Abdaljalil et al. (2026) studied abstention-aware scientific reasoning by proposing a verification framework that decomposes claims into conditions, audits them via natural language inference (NLI), and selectively decides to support, refute, or abstain (Abdaljalil et al., 2026, 5). A key strength of the study is its comprehensive evaluation across SciFact and PubMedQA using diverse models and risk-coverage analysis, though it is limited by its reliance on an NLI model to audit conditions (Abdaljalil et al., 2026, 5).

Complementing verification pipelines with a focus on retrieval dynamics, Zhou et al. (2025) investigated the refusal capabilities and uncertainty calibration of retrieval-augmented language models (RALMs), demonstrating that while models achieve good calibration when supportive documents are present, they exhibit a strong over-refusal tendency when confronted exclusively with irrelevant contexts (Zhou et al., 2026, 6). A key strength of the study is its comprehensive evaluation framework combining multiple uncertainty estimation methods and refusal instruction-tuning techniques across diverse knowledge states, though its primary limitation is its focus on simple factual question-answering benchmarks rather than complex decision-making scenarios (Zhou et al., 2026, 6).

Collectively, these studies highlight a critical research gap concerning how evidence ordering and conflicting contexts specifically influence model reliability and answer-abstention behavior. To advance beyond these static and QA-focused evaluations, future research could improve upon existing studies by applying alternative statistical methods or advanced machine-learning techniques to predict complex domain-specific outcomes, such as women’s empowerment metrics, under conditions of conflicting or uncertain evidence.